

Involution Pseudoknot-bordered Words *

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Abstract

This paper aims to investigate properties of involution pseudoknot words. Let θ be either an antimorphic involution. We give a characterization that $L_d^\theta(w) = L_{cd}^\theta(w)$ for a word w , where $L_d^\theta(w)$ denote the set of all proper θ -borders of a nonempty word w and $L_{cd}^\theta(w)$ denote the set of all θ -pk-borders of a nonempty word w . We show that some properties of θ -pk-unbordered words. There exist nonprimitive θ -pk-unbordered words. The root of these words are also θ -pk-unbordered words. We also prove that every nonempty skew θ -palindrome word is θ -pk-bordered.

Keywords: *Involution Palindrome, Skew Involution Palindrome, Involution Pseudoknot-Unbordered Words, Involution Bordered Words.*

1 Introduction

Theoretical DNA Computing is an area of biomolecular computing that loosely encompasses contributions to fundamental research in computer science originated in or motivated by research in DNA computing. One of the most active areas of research in theoretical DNA computing is the search for ways to encode information on DNA for the purposes of biocomputation that ensure that no unwanted bindings occur.

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The main premise is that information-encoding strings that are used in DNA computing experiments have an important property that differentiates them from their electronic computing counterparts. This property is the Watson-Crick complementarity between DNA single-strands that allows information-encoding strands to potentially interact.

A DNA single strand is a linear chain made up of four different types of nucleotides, each consisting of a sugar-phosphate unit and a base (Adenine, Cytosine, Guanine or Thymine). The sugar-phosphate units are linked together by strong covalent bonds, to form the backbone of the DNA strand. Since nucleotides may differ only by their bases, a DNA single strand can be viewed as a string over the DNA alphabet of bases $\{A, C, G, T\}$. A DNA single strand has an orientation, with one end known as the 3'-end, and the other known as the 5'-end, based on their chemical properties. By convention, a word over the DNA alphabet represents a DNA single strand in its 5' to 3' orientation. An essential biochemical property of DNA single strands is that of Watson-Crick complementarity, wherein A can bind to T , and C can bind to G by weak hydrogen bonds. Two Watson-Crick complementary DNA single strands of opposite orientation can bind to each other to form a DNA double strand. This and other biochemical properties of DNA have all been harnessed in biomolecular computing, in which information is encoded as DNA single strands, and processed through bio-operations.

Most DNA-based computations consist of three

basic stages. The first is encoding the input data using DNA stranded molecules, the second is performing the biocomputation using bio-operations and the third is decoding the result. One of the main problems associated with such biocomputations is the design of the information-encoding oligonucleotides such that undesirable pairing due to the Watson-Crick complementarity is minimized. There are several approaches exist that address this sequence design problem. Such as the software simulation approach, the algorithmic approach and the theoretical approach to the design of optimal data-encoding DNA strands.

Theoretical DNA computing has been an enthusiastic research area in computer science for a decade. Studies include the computational properties of DNA recombination ([2]), the mathematical theory of DNA self-assembly ([1]), and the coding properties of DNA languages ([4]) has inspired the authors to study a similar subject. This study focus on a theoretical study of generalized notions of palindrome words. The motivation comes from the DNA encoded sequences which have an important property used in DNA computing. This property is the Watson-Crick complementarity between DNA single-strands. A single strand of DNA can be abstracted as a string sequence consisted of a combination of four different symbols, known as A, C, G, T . The Watson-Crick complement of a DNA strand satisfies two main properties, it is the reverse property and the complement property of the original strand. Mathematically, the Watson-Crick complementarity is translated into generalizing the identity involution. An involution is a function θ such that θ^2 equals the identity, that is, $\theta(A) = T$ and $\theta^2(A) = \theta(\theta(A)) = A$. Furthermore, for the DNA alphabet $X = \{A, G, C, T\}$, an antimorphic involution is an involution θ with the additional property that $\theta(uv) = \theta(v)\theta(u)$ for all string sequences $u, v \in X^*$, corresponds to the notion of the Watson-Crick complement of a DNA sequence.

2 Preliminaries

Assume X is an alphabet containing more than one letter. Let X^* be the free monoid generated by X . Every element of X^* is a *word* and every subset of X^* is a *language*. Let λ denote the empty word, and $X^+ = X^* \setminus \{\lambda\}$. For $w \in X^*$ and $L \subseteq X^*$, let $\lg(w)$ denote the length of the word w and let $|L|$ denote the cardinality of the language L . A language $L \subseteq X^*$ is *dense* if for any $w \in X^*$, there

exist $x, y \in X^*$ such that $xwy \in L$. That is, for every $w \in X^*$, $X^*wX^* \cap L \neq \emptyset$. A *primitive word* is a word which is not a power of any other word. Let Q be the set of all primitive words over X . Every word $u \in X^+$ can be expressed as a power of a primitive word in a unique way, that is, for any $u \in X^+$, $u = f^n$ for a unique $f \in Q$ and $n \geq 1$. In this case, f is the primitive root of u and denoted by $\sqrt{u} = f$. Let $u = a_1a_2 \cdots a_n$ where $a_i \in X$. The reverse of the word u is $u^R = a_n \cdots a_2a_1$. A word u is called *palindrome* if $u = u^R$. Let R be the set of all palindrome words over X . The partial order relation $\leq_p$ (resp. $\leq_s$) is defined as: for $u, v \in X^*$, $v \leq_p u$ (resp. $v \leq_s u$) if and only if $u \in vX^*$ (resp. $u \in X^*v$). Moreover, the partial order relation $<_p$ (resp. $<_s$) is defined as: for $u, v \in X^+$, $v <_p u$ (resp. $v <_s u$) if and only if $u \in vX^+$ (resp. $u \in X^+v$).

Let $\theta : X^* \rightarrow X^*$ be a function such that $\theta^2 = I$ where I is the identity function. It can extend to a morphic involution on X^* if for all $u, v \in X^*$, $\theta(uv) = \theta(u)\theta(v)$ or an antimorphic involution if $\theta(uv) = \theta(v)\theta(u)$. We now recall some definitions introduced by Kari and Mahalingam ([5]-[7]).

Definition 2.1 Let θ be either a morphic or an antimorphic involution on X^* .

1. A word w is a θ -conjugate of another word u if $uv = \theta(v)w$ for some $v \in X^*$.
2. A word $w \in X^*$ is called a θ -palindrome word if $w = \theta(w)$.

Definition 2.2 Let θ be an antimorphic involution on X^* . A word w is said to be *skew θ -palindrome* if $w = xy$ implies that $\theta(w) = yx$ for some $x, y \in X^*$.

Let θ is an antimorphic involution on X^* . Note that every θ -palindrome word is a skew θ -palindrome word because any palindrome word as a product of itself and the empty word λ . For example, $abab$ is θ -palindrome and is also a skew θ -palindrome word. Let $w = abba$. Then w is a skew θ -palindrome word, but it is not θ -palindrome.

Furthermore, we give some results which will be used in the rest of this study as follows:

Lemma 2.1(see [13]) Let $u, v \in X^+$. Then $uv = vu$ implies that u and v are powers of a common word.

3 θ -pseudoknot-bordered words

In this section we propose the notion of θ -pseudoknot-bordered words for a morphic or antimorphic (or d-morphic) involution θ . If we consider the DNA alphabet $\{A, C, G, T\}$, wherein θ is the Watson-Crick complementarity function, then a word that is θ -pseudoknot-unbordered will not form pseudoknot-like secondary structures. The notion of θ -pseudoknot-bordered word is a proper generalization of the notion of θ -bordered word proposed in [11], and thus a proper generalization of the notion of bordered word. Let θ be a morphic involution or an antimorphic involution. A word $u \in X^+$ is said to be a proper θ -border of a word $w \in X^+$ if u is a proper prefix of w and $\theta(u)$ is a proper suffix of w , i.e., $w = u\alpha = \beta\theta(u)$ for some $\alpha, \beta \in X^+$. Let $L_d^\theta(w)$ denote the set of all proper θ -borders of a nonempty word w . Note that $\lambda \in L_d^\theta(w)$ for all $w \in X^+$. A word $w \in X^+$ is said to be θ -bordered if it has a proper θ -border other than λ , i.e., $|L_d^\theta(w)| \geq 2$; otherwise, it is θ -unbordered. Define now $D_\theta(i) = \{w \in X^+ \mid |L_d^\theta(w)| = i\}$ for every $i \geq 1$. Then $D_\theta(1)$ is the set of all θ -unbordered words.

We call a word $u \in X^*$ a θ -pseudoknot-border (or θ -pk-border) of a word $w \in X^*$ if there exists a cyclic permutation v of u such that $w = u\alpha = \beta\theta(v)$ for some $\alpha, \beta \in X^*$. We also employ the expression " xy is a θ -pk-border of w " to mean " v is a θ -pk-border of w such that $v = xy$ and $w = xy\alpha = \beta\theta(yx)$ for some $\alpha, \beta \in X^*$." Let $L_{cd}^\theta(w)$ denote the set of all θ -pk-borders of a nonempty word w , and $K_\theta(i) = \{w \in X^+ \mid |L_{cd}^\theta(w)| = i\}$ for every $i \geq 1$. We call a nonempty word θ -pseudoknot-bordered (or θ -pk-bordered) if it has a nonempty θ -pk-border; otherwise, it is θ -pseudoknot-unbordered. Note that $\lambda \in L_{cd}^\theta(w)$ for all $w \in X^+$. Note also that no word in $K_\theta(1)$ has θ -pk-borders other than λ , and hence $K_\theta(1)$ is the set of all θ -pk-unbordered words.

Lemma 3.1. ([14]) Let θ be a d-morphic involution on X^* and $w \in X^+$. Then $L_d^\theta(w) \subseteq L_{cd}^\theta(w)$.

Lemma 3.2. ([14]) Let θ be a d-morphic involution on X^* . Then $K_\theta(1) \subseteq D_\theta(1)$.

Example 1. Let $X = \{a, b\}$ be an alphabet. Let θ be an antimorphic involution on X^* with $\theta(a) = b$ and $\theta(b) = a$. Let $w_1 = aabb$, $w_2 = aab$, $w_3 = aa$, and $w_4 = aba$. Then

1. $L_d^\theta(w_1) = \{\lambda, a, aa, aab\}$ and $L_{cd}^\theta(w_1) = \{\lambda, a, aa, aab, aabb\}$. Thus, $L_d^\theta(w_1) \subset L_{cd}^\theta(w_1)$. And $w_1 \in D_\theta(4)$, $w_1 \in K_\theta(5)$.
2. $L_d^\theta(w_2) = \{\lambda, a\}$ and $L_{cd}^\theta(w_2) = \{\lambda, a\}$. Thus, $L_d^\theta(w_2) = L_{cd}^\theta(w_2)$. And $w_2 \in D_\theta(2)$, $w_2 \in K_\theta(2)$.
3. $L_d^\theta(w_3) = \{\lambda\}$ and $L_{cd}^\theta(w_3) = \{\lambda\}$. Thus, $L_d^\theta(w_3) = L_{cd}^\theta(w_3)$. And $w_3 \in D_\theta(1)$, $w_3 \in K_\theta(1)$.
4. $L_d^\theta(w_4) = \{\lambda\}$ and $L_{cd}^\theta(w_4) = \{\lambda, ab\}$. Thus, $L_d^\theta(w_4) \subset L_{cd}^\theta(w_4)$. And $w_4 \in D_\theta(1)$, $w_4 \in K_\theta(2)$, that is, $w_4 \notin K_\theta(1)$.

From above examples, we see that $L_d^\theta(w)$ may equal to $L_{cd}^\theta(w)$. The following lemmata are clear:

Lemma 3.3 Let θ be a d-morphic involution on X^* . Then $L_d^\theta(w) = L_{cd}^\theta(w)$ for every $w \in D_\theta(1) \cap K_\theta(1)$. ■

Lemma 3.4 Let $X = \{a, b\}$ be an alphabet. Let θ be a d-morphic involution on X^* with $\theta(a) = b$, $\theta(b) = a$ and let $w \in X^+$. Then $L_d^\theta(w) = L_{cd}^\theta(w)$ for every $x, y \in X^*$ with $xy \leq_p w$ and $\theta(yx) \not\leq_s w$. ■

In Proposition 3.6, we give a characterization for $L_d^\theta(w) = L_{cd}^\theta(w)$, where θ is a d-morphic involution.

Lemma 3.5. ([14]) Let θ be a d-morphic involution on X^* . The following hold:

1. If a word $w \in X^+$ has a θ -pk-border of length n , then, for every $a \in X$, the number of occurrences of the letter a in the prefix of length n of w is equal to the number of occurrences of the letter $\theta(a)$ in the suffix of length n of w .
2. For all $a \in X$ such that $a \neq \theta(a)$, a^k is θ -pk-unbordered for all $k \geq 1$.
3. For words $v, w \in X^+$ and $n \geq 1$, if $v \in L_{cd}^\theta(w^n)$ and $|w^{m-1}| < |v| \leq |w^m|$ for some $m \geq 1$, then $v \in L_{cd}^\theta(w^k)$ for all k with $m \leq k \leq n$.

For a word $w \in X^*$, a word $v \in X^*$ is called a *cyclic permutation (or conjugate)* of w if there exist two words $x, y \in X^*$ such that $w = xy$ and

$v = yx$. We denote the set of all cyclic permutations (or conjugates) of w by $Cp(w)$, that is, $Cp(w) = \{yx | w = xy, x, y \in X^*\}$. From above Lemma 3.5, we have the following proposition:

Proposition 3.6 Let θ be a d-morphic involution on X^* . Let $w \in X^n$ where $n \geq 1$. Then $L_d^\theta(w) = L_{cd}^\theta(w)$ if and only if one of the following conditions holds:

1. $w \in K_\theta(1)$.
2. $w \notin L_{cd}^\theta(w)$ and $\theta(Cp(Pref(w))) \cap Suff(w) \subseteq \theta(Pref(w)) \cap Suff(w)$.

Proof. Let θ be a d-morphic involution on X^* . Let $w \in X^n$ where $n \geq 1$.

($\Rightarrow$) Assume that $L_d^\theta(w) = L_{cd}^\theta(w)$. If $L_d^\theta(w) = L_{cd}^\theta(w) = \{\lambda\}$, then $w \in D_\theta(1)$ and $w \in K_\theta(1)$. Thus (1) holds. If $L_d^\theta(w) = L_{cd}^\theta(w) \neq \{\lambda\}$, Thus there exists a word $u \in X^+$ such that all the following conditions are true.

- $u\alpha_1 = \beta_1\theta(u)$ for some $\alpha_1, \beta_1 \in X^+$.
- $xy\alpha_2 = \beta_2\theta(yx)$ for some $x, y \in X^*$ with $xy \neq \lambda$ and $\alpha_2, \beta_2 \in X^*$.

Since $L_d^\theta(w) = L_{cd}^\theta(w)$, this in conjunction with $w \notin L_d^\theta(w)$, we get that $w \notin L_{cd}^\theta(w)$. Now there exist a word $u \in X^+$ and an integer $1 \leq m < |w|$ such that $u \in L_d^\theta(w) = L_{cd}^\theta(w)$ with $|u| = m$. From the definition of θ -border, there exists at least one θ -border with length $m \geq 0$. This in conjunction with $L_d^\theta(w) = L_{cd}^\theta(w)$, we get that (2) holds.

($\Leftarrow$) Assume that Conditions (1) and (2) hold. Suppose that $L_d^\theta(w) \neq L_{cd}^\theta(w)$. By Lemma 3.1, there exists a word $u \in L_{cd}^\theta(w) \setminus L_d^\theta(w)$. Thus there exists a nonempty θ -pk-border u such that $u = xy$ for some $x, y \in X^*$ with $xy \neq \lambda$. Thus $w = xy\alpha = \beta\theta(yx)$. This implies that $w \notin K_\theta(1)$, this contradicts Condition (1). From $w = xy\alpha = \beta\theta(yx)$ for some $x, y \in X^+$, this implies that $\theta(Cp(Pref(w))) \cap Suff(w) \not\subseteq \theta(Pref(w)) \cap Suff(w)$. This contradicts to Condition (2). Hence $L_d^\theta(w) = L_{cd}^\theta(w)$. ■

4 θ -Palindrome and θ -pk-bordered Words

A word u is called palindrome if it is the mirror image of itself. In this section we extend the concept of palindrome words to incorporate the

notion of involution function. The notion of θ -palindrome was defined in [6] and obtained independently in [8]. Note that if θ is the Watson-Crick involution, then the notion of Watson-Crick palindromes coincides with the term "palindrome" as used in molecular biology, especially in the study of enzymes.

The study of θ -palindromes for antimorphic involutions is interesting from two points of view: firstly, it may be desirable for certain DNA computing experiments to use DNA strands that contain θ -palindromic enzyme restriction sites as subwords, and secondly, in general, a set of DNA codewords should be free of θ -palindromic words, due to the intermolecular hybridizations that these would entail.

This study extends the understanding of involution palindrome words (θ -palindrome words). This notion is motivated by DNA strand design in the area of biocomputing where the Watson-Crick complementarity can be abstracted as an antimorphic involution function. A language consisting of involution palindrome words is defined as an involution palindrome language. Besides involution palindrome words being considered, some algebraic properties of skew involution palindrome words are studied as well. We also investigate the relation between θ -palindrome words and θ -pk-bordered words. The following is known result.

Lemma 4.1. ([14]) Let θ be an antimorphic involution on X^* , and x, y be θ -palindromes such that $xy \neq \lambda$. If a word $u \in X^+$ has xy as both its prefix and suffix, then u is θ -pk-bordered.

Lemma 4.2 ([15]) Let θ be an antimorphic involution on X^* . A word w is skew θ -palindrome if and only if w is a product of two θ -palindrome words.

From the above lemmata, the following is clear:

Lemma 4.3. Let θ be an antimorphic involution on X^* . Then every nonempty skew θ -palindrome word is θ -pk-bordered.

Proof. Let $w \in X^+$ is a skew θ -palindrome word. By Lemma 4.2, there exist x, y are θ -palindrome words such that $w = xy$. That is, xy is a prefix of w and also a suffix of w . This in conjunction with x, y are θ -palindrome, by Lemma 4.1, we have that w is θ -pk-bordered. ■

From Lemma 4.3 and Proposition 3.6, we now have the following corollary:

Corollary 4.4. Let θ be an antimorphic involution on X^* . If w is a skew θ -palindrome word, then $L_d^\theta(w) \neq L_{cd}^\theta(w)$. ■

Since a θ -palindrome word is also a skew θ -palindrome word, the following also true:

Corollary 4.5. Let θ be an antimorphic involution on X^* . If w is a θ -palindrome word, then $L_d^\theta(w) \neq L_{cd}^\theta(w)$. ■

Let $w \notin Q$. That is, $w = f^k, f \in Q, k \geq 2$. Then the primitive root of w is denoted by $\sqrt{w} = f$.

Lemma 4.6. ([7]) Let θ be a morphic or an antimorphic involution. For all $w \in X^+$, w is θ -palindrome if and only if $\sqrt{w}$ is θ -palindrome.

In the case of θ -pk-bordered words, the above result is not true. For example, let θ be an antimorphic involution and let $w = aabbbbaba$. Then $u \in K_\theta(w)$ and $u^2 \notin K_\theta(w)$. That is, u is a θ -pk-unbordered word but u^2 is a θ -pk-bordered word. This implies that $\sqrt{w}$ may be a θ -pk-unbordered word for a θ -pk-bordered word w . In the following, we will show that for every θ -pk-unbordered word w , we have that $\sqrt{w}$ is also a θ -pk-unbordered. First, we give a nonprimitive θ -pk-unbordered word. Let $w = a^2$, where $a \in X$. It is clear that $w, \sqrt{w}$ are θ -pk-unbordered words.

Proposition 4.7. Let θ be a morphic or an antimorphic involution. Then $\sqrt{w} \in K_\theta(1)$ for every $w \in K_\theta(1)$.

Proof. Let θ be a morphic or an antimorphic involution and $w \in X^+$. If $w \in Q \cap K_\theta(1)$, then the result is immediate. Now we consider $w \in K_\theta(1) \subseteq Q$. Since $w \in K_\theta(1)$, w has no θ -pk-border with length $m \geq 1$. That is, it has no θ -pk-border with length $m \leq |w|$. This implies that $w \in K_\theta(1)$. ■

5 Conclusion

This paper studies algebraic properties concerning involution pseudoknot words. Let θ be either a morphic involution or an antimorphic involution. We give a characterization that $L_d^\theta(w) = L_{cd}^\theta(w)$ for a word w . We show that some properties of θ -pk-unbordered words. There exist nonprimitive θ -pk-unbordered words. The root of these words are also θ -pk-unbordered words. We also prove

that every nonempty skew θ -palindrome word is θ -pk-bordered.

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